Operator-system extension of the BBB Krein spectral triple at finite cutoff: propagation number and envelope dependence on the Camporesi–Higuchi spinor bundle

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Abstract

The Connes–van Suijlekom [6] spectral truncation of $C^\infty(S^3)$ on the round $S^3 = \mathrm{SU}(2)$ at cutoff $n_{\max}$ produces a $*$-closed but *not* multiplicatively closed linear subspace $\mathcal{O}_{n_{\max}} \subset M_N(\mathbb{C})$ with propagation number $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ (Paper 32 § III, matching the Connes–vS worked example on the Toeplitz operator system $C(S^1)^{(n)}$ verbatim). We construct the natural Lorentzian extension $\mathcal{O}^L_{n_{\max},N_t} \subset \mathcal{B}(\mathcal{K}_{n_{\max},N_t})$ on the Bizi–Brouder–Besnard [3] Krein spectral triple at signature $(s,t) = (3,1)$, $(m,n) = (4,6)$, in the Peskin–Schroeder chiral basis with Lorentzian Dirac $D_L = i\big(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}\big)$ per van den Dungen 2016 [26] Proposition 4.1, and the Camporesi–Higuchi [5] spatial spinor bundle. Multipliers are tensor products of chirality-doubled spatial scalar 3-Y multipliers with a polynomial-basis temporal subalgebra on a finite uniform t-grid. At $N_t = 1$ the construction reduces bit-identically to the Riemannian chirality-doubled operator system on the FullDirac Hilbert space, with Frobenius residual exactly 0.0 in float64. We prove $\mathcal{O}^L_{n_{\max},N_t}$ is a Lorentzian truncated operator system (it is $*$-closed, contains the identity, and is not multiplicatively closed) and exhibit a witness pair lifting the Paper 32 § III witness verbatim. The principal structural finding is that the propagation number is *envelope-dependent*: targeting the natural Weyl-doubled achievable envelope $\dim^2_{\mathrm{Weyl}} \cdot N_t$, we obtain $\mathrm{prop}(\mathcal{O}^L_{n_{\max},N_t}) = 2$ for every $n_{\max} \geq 2$, matching Paper 32 § III verbatim; targeting the full $\mathcal{B}(\mathcal{K})$ envelope $\dim^2_{\mathcal{K}}$, we obtain $\mathrm{prop} = \infty$ at every cell. The Weyl-doubling and the commutative temporal subalgebra *block* scalar multipliers from generating chirality-flipping or non-diagonal temporal operators; the achievable envelope is the correct natural target for scalar-multiplier closure on the spinor bundle. A separate substantive finding is that the Krein-positive restriction is trivial: every chirality-doubled scalar multiplier $M \oplus M$ commutes with the chirality-swap fundamental symmetry J_L, so $\mathcal{O}^{L,+} = \mathcal{O}^L$ exactly. The Krein-positive program is shifted from the operator-system substrate (where it is automatic) to the state level, which is the natural setting for any future Lorentzian-propinquity convergence argument. The construction is the foundation on which any subsequent continuum Lorentzian-propinquity program (Sprint L3b, see § 9) would build. Honest scope: this paper closes the operator-system substrate at finite cutoff; continuum / GH-limit / Lorentzian-propinquity convergence is open. No published Lorentzian propinquity construction exists as of May 2026: Latrémolière 2017/2023 [15] and Hekkelman–McDonald 2024 [13] are strictly Riemannian.

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Keywords: Connes–van Suijlekom truncated operator system, propagation number, Krein spectral triple, Bizi–Brouder–Besnard indefinite signature, Camporesi–Higuchi Dirac, Lorentzian noncommutative geometry, spectral truncation, finite-cutoff operator-system substrate.

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1 Introduction

A central technical object in the spectral-truncation framework of Connes and van Suijlekom [6] is the truncated operator system

$$\mathcal{O}_\Lambda \;=\; P_\Lambda\, C^\infty(M)\, P_\Lambda \;\subset\; \mathcal{B}(\mathcal{H}_\Lambda), \tag{1}$$

where P_Λ projects $L^2(M)$ onto the span of the first Λ eigenfunctions of the Dirac operator (or, equivalently in the GeoVac convention, onto the first $n_{\max}$ Fock shells of the Fock-projected spectrum). On the round $S^3 = \mathrm{SU}(2)$ with spinor-harmonic basis adapted to the SO(4) decomposition, this construction was carried out in Paper 32 § III of the present author's framework documentation and verified to have propagation number

$$\mathrm{prop}(\mathcal{O}_{n_{\max}}) \;=\; 2 \qquad (n_{\max} \in \{2, 3, 4\}), \tag{2}$$

matching the Connes–vS worked example on the Toeplitz operator system $C(S^1)^{(n)}$ verbatim [6]. This was the strongest single quantitative alignment between GeoVac and the published Connes–vS framework as of the Round-2 sprint (`debug/wh1_round2_propagation_memo.md`, 2026-05-03).

The companion Paper 43 [33] of the present author closes the four-witness Wick-rotation theorem (Hartle–Hawking [11], Sewell [22], Bisognano–Wichmann [2], Unruh [25]) at the operator-system level on the Lorentzian Krein spectral triple at signature $(3, 1)$ in the Bizi–Brouder–Besnard [3] classification at $(m, n) = (4, 6)$, with the modular Hamiltonian and Tomita structure on a hemispheric wedge. Paper 43 takes the operator-system substrate as input; the present paper constructs it.

1.1 What this paper does

We define and study the natural Lorentzian extension

$$\mathcal{O}^L_{n_{\max},N_t} \;=\; P_{n_{\max},N_t} \cdot \left[\, C^\infty(S^3) \otimes C^\infty(\mathbb{R}_t)_{\mathrm{cutoff}} \,\right] \cdot P_{n_{\max},N_t} \;\subset\; \mathcal{B}(\mathcal{K}_{n_{\max},N_t}) \tag{3}$$

on the Krein space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}^{n_{\max}}_{\mathrm{GV}} \otimes \mathbb{C}^{N_t}$ of Paper 43 § 2 (Sprint L2-B). Here $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$ is the chirality-doubled Camporesi–Higuchi [5] spinor space at Fock cutoff $n_{\max}$, with $\dim_{\mathbb{C}} \mathcal{H}^{n_{\max}}_{\mathrm{GV}} = \frac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2) \in \{4, 16, 40, 80, 140\}$ at $n_{\max} \in \{1, 2, 3, 4, 5\}$, and $\mathbb{C}^{N_t}$ is the temporal L^2 cutoff on the uniform grid $t_k = -T_{\max} + k \cdot 2T_{\max}/(N_t - 1)$. Spatial scalar multipliers act on $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$ via the chirality-block-doubled spinor lift (Clebsch–Gordan decomposition $\oplus$ Avery–Wen–Avery 3-Y integrals); temporal multipliers act on $\mathbb{C}^{N_t}$ by pointwise multiplication via the polynomial basis $g_p(t) = t^p$ for $p = 0, \ldots, N_t - 1$.

The principal results are:

1. **Operator-system axioms hold** (Proposition 3.2, Section 3). $\mathcal{O}^L_{n_{\max},N_t}$ contains the identity, is closed under conjugate transposition, and is *not* multiplicatively closed: an explicit witness pair lifting the Paper 32 spatial witness produces a product ab with Frobenius residual $\sim 38\%$ relative to the linear span of $\mathcal{O}^L$ (Theorem 7.1, Section 7).

2. **Riemannian limit at $N_t = 1$ is bit-exact** (Theorem 4.1, Section 4). The construction reduces to the chirality-doubled FullDirac operator system on $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$ with maximum residual 0.0 in float64 at every $n_{\max} \in \{1, 2, 3\}$, verifying compatibility with the Sprint L2-B Krein-space basis convention.

3. **Envelope-dependent propagation number** (Theorem 5.2, Section 5). Targeting the Weyl-doubled *achievable* envelope $\dim_{\mathrm{ach}} = \dim^2_{\mathrm{Weyl}} \cdot N_t$,

$$\mathrm{prop}_{\mathrm{ach}}(\mathcal{O}^L_{n_{\max},N_t}) \;=\; 2 \qquad \text{for every } n_{\max} \geq 2,\; N_t \geq 1, \tag{4}$$

verbatim with Paper 32 § III. Targeting the full $\mathcal{B}(\mathcal{K})$ envelope $\dim_{\mathcal{K}}^2$,

$$\mathrm{prop}_{\mathrm{full}}(\mathcal{O}^L_{n_{\max},N_t}) \;=\; \infty \qquad \text{at every cell.} \tag{5}$$

The discrepancy is a real structural feature of the chirality-doubled spinor bundle: scalar multipliers $M \oplus M$ cannot generate chirality-flipping operators (off-diagonal blocks of the chirality splitting), and a commutative temporal subalgebra cannot generate non-diagonal temporal operators. Eq. (4) is the natural Connes–vS-style propagation question for scalar multipliers on a chirality-doubled bundle; (5) is a strictly larger naive envelope not reached by the scalar-multiplier construction.

4. **Krein-positive restriction is trivial** (Proposition 6.1, Section 6). Every multiplier in $\mathcal{O}^L_{n_{\max},N_t}$ commutes with the chirality-swap fundamental symmetry $J_L = \gamma^0 \otimes I_{N_t}$, so it preserves the Krein-positive cone $\mathcal{K}^+ = \ker(J_L - I)$. The natural restriction $\mathcal{O}^{L,+}$ to Krein-positive preservers equals $\mathcal{O}^L$ itself.

1.2 Position in the literature

The Lorentzian-propinquity question sits at the intersection of three threads with empty intersection as of May 2026:

(a) **Riemannian propinquity convergence on spectral truncations.** Connes–van Suijlekom [6] introduces the truncated operator system and the propagation-number invariant; Latrémolière [15, 16] develops the propinquity for metric spectral triples; the explicit Gromov–Hausdorff convergence on a non-trivial example was deferred to “elsewhere” three times in [6] (lines 220–221, 1101, 2616–2617 of arXiv:2004.14115v2). Subsequent work by Hekkelman [12], Hekkelman–McDonald [13, 14], Leimbach–van Suijlekom [17], Toyota [24], Farsi–Latrémolière [8, 9] addresses the flat structures S^1, T^d, and Berezin–Toeplitz S^2. The first non-abelian case $S^3 = \mathrm{SU}(2)$ was closed by the present author in Paper 38 [29]; the tensor-product extension was closed in Paper 39 [30]; the universal compact-Lie-group rate constant $4/\pi$ was closed in Paper 40 [31]. All of these are strictly Riemannian.

(b) **Lorentzian noncommutative geometry.** Strohmaier [23], van den Dungen [26], Franco–Eckstein [10], Bizi–Brouder–Besnard [3], Devastato–Lizzi–Martinetti [7], and Nieuviarts [19] construct indefinite (Krein) spectral triples and address Wick rotation at the level of the data $(\mathcal{A}, \mathcal{K}, D_L, J_L)$. None of these works carries the construction through to a Connes–van Suijlekom truncated operator system on the Krein side, nor to a propagation-number invariant.

(c) **Lorentzian Gromov–Hausdorff convergence.** The recent Mondino–Sämann program on synthetic Lorentzian GH [18] and the Bykov–Minguzzi–Suhr work on Lorentzian convergence of metric spaces [4] develop a metric-space-side notion of Lorentzian convergence but do not interface with the operator-system/spectral-triple side.

The present paper closes the operator-system *substrate* of any prospective Lorentzian-propinquity convergence program at finite cutoff; it does not close the propinquity itself. The substrate is the foundation on which a future propinquity argument would be built.

1.3 Why this is a separate paper from Paper 43

Paper 43 [33] closes the four-witness Wick-rotation theorem at the *modular-Hamiltonian* level on a hemispheric wedge of the Lorentzian Krein space, using the same Krein construction $\mathcal{K}_{n_{\max},N_t}$, Lorentzian Dirac D_L, and BBB axiom audit at $(m,n) = (4,6)$ as the present paper. The Paper 43 results (integer spectrum of the geometric BW-α generator, β-independent wedge KMS state, Tomita–Takesaki polar decomposition, flow conjugacy, six-witness collapse) operate at the level of *specific operators* on the Krein space: the wedge projector, the modular operator Δ, the modular Hamiltonian K_L^α and K_L^{TT}. They do not directly study the linear span of all spatial-times-temporal multipliers as a $*$-closed subspace of $\mathcal{B}(\mathcal{K})$.

The present paper studies that subspace explicitly. The propagation-number invariant is independent of any wedge structure or modular flow: it is a statement about the multiplicative closure of the operator system as a whole, and its envelope-dependence at the Weyl-doubled vs full $\mathcal{B}(\mathcal{K})$ level is a separate structural feature not visible in Paper 43's hemispheric-wedge analysis. Conceptually, Paper 43 is the analog of computing modular data on a specific algebra (here the wedge-restricted operator system); the present paper is the analog of constructing the algebra itself together with its envelope-relative structural invariants. Computationally, the two papers share their lower layers (`geovac/krein_space_construction.py`, `geovac/lorentzian_dirac.py`, `geovac/connes_axiom_audit_31.py`) but Paper 43 does not import `geovac/operator_system_lorentzian.py` (the Sprint L3a-1 module) and vice versa.

1.4 Outline

Section 2 recalls the Krein-space construction at $(m,n)=(4,6)$ from Paper 43 and fixes notation. Section 3 defines the Lorentzian truncated operator system $\mathcal{O}^L$, proves $*$-closure and identity-membership, and discusses the multiplier bookkeeping. Section 4 verifies the Riemannian-limit recovery at $N_t=1$ (load-bearing falsifier). Section 5 proves the envelope-dependent propagation theorem. Section 6 establishes the trivial Krein-positive restriction. Section 7 exhibits the witness pair. Section 8 connects to the Paper 43 hemispheric wedge. Section 9 states open questions and the recommended first Sprint L3b move. Section 10 concludes.

2 Setup: Krein space, Lorentzian Dirac, BBB axiom audit

Sections 2.1–2.3 recall the data of the Lorentzian Krein spectral triple from Paper 43 [33] § 3 (Sprints L2-B/C/D). The reader familiar with Paper 43 may skim or skip these subsections.

2.1 The Krein space $\mathcal{K}_{n_{\max},N_t}$

Fix a Fock cutoff $n_{\max}\in\mathbb{N}$ and a temporal grid size $N_t\in\mathbb{N}$ on a half-width $T_{\max}>0$. The Krein space at cutoff is

$$\mathcal{K}_{n_{\max},N_t} \;:=\; \mathcal{H}_{\mathrm{GV}}^{n_{\max}}\otimes\mathbb{C}^{N_t}, \tag{6}$$

where $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ is the chirality-doubled Camporesi–Higuchi spinor space at cutoff $n_{\max}$ (Paper 32 § III [28], Paper 43 § 3.1) with basis labelled $\big(n_{\mathrm{fock}}, l, 2m_j, \mathrm{chirality}\big)$ in the `FullDiracLabel` convention. The spatial dimension is

$$\dim_{\mathbb{C}}\mathcal{H}_{\mathrm{GV}}^{n_{\max}} \;=\; \tfrac{2}{3}\,n_{\max}(n_{\max}+1)(n_{\max}+2) \;=\; \begin{cases} 4 & n_{\max}=1\\ 16 & n_{\max}=2\\ 40 & n_{\max}=3\\ 80 & n_{\max}=4\\ 140 & n_{\max}=5\end{cases} \tag{7}$$

and the total Krein dimension is $\dim_{\mathbb{C}}\mathcal{K}_{n_{\max},N_t}=\dim_{\mathbb{C}}\mathcal{H}_{\mathrm{GV}}^{n_{\max}}\cdot N_t$.

We use the symmetric temporal grid

$$t_k \;:=\; -T_{\max}+k\cdot\tfrac{2T_{\max}}{N_t-1} \qquad (k=0,\dots,N_t-1), \tag{8}$$

with the convention $t_0=0$ at $N_t=1$. The fundamental symmetry on $\mathcal{K}_{n_{\max},N_t}$ is

$$J_L \;:=\; \gamma^0\otimes I_{N_t}, \tag{9}$$

where γ^0 acts as chirality-swap on $\mathcal{H}_{\rm GV}$ in the Peskin–Schroeder chiral basis (West-coast metric $\eta = \mathrm{diag}(+,-,-,-)$). The basic Krein-space axioms $J_L^2 = +I$, $J_L^\dagger J_L = I$, $J_L^\dagger = J_L$, and the splitting $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$ into ± 1-eigenspaces of J_L are verified bit-exactly in Sprint L2-B (Paper 43 § 3.1).

2.2 The Lorentzian Dirac D_L

The Lorentzian Dirac operator on $\mathcal{K}_{n_{\rm max},N_t}$ is

$$D_L \;:=\; i\big(\gamma^0 \otimes \partial_t \;+\; D_{\rm GV} \otimes I_{N_t}\big), \tag{10}$$

per van den Dungen 2016 [26] Proposition 4.1 with $(s,t) = (3,1)$ giving $i^t = i^1 = i$, where $D_{\rm GV}$ is the truthful Camporesi–Higuchi Dirac on $\mathcal{H}_{\rm GV}^{n_{\rm max}}$ (spectrum $|\lambda_n| = n + \frac{1}{2}$ with multiplicities $g_n^{\rm Dirac} = 2(n+1)(n+2)$ on each chirality) and ∂_t is the centered finite-difference operator on the uniform grid (8) with Dirichlet zero boundary conditions. The sign $i^t = +i$ is structurally forced by Krein-self-adjointness $D_L^\times = D_L$ (Paper 43 § 3.2, `geovac/lorentzian_dirac.py`). At $N_t = 1$, $\partial_t = 0$ and $D_L = iD_{\rm GV}$ reduces to the Riemannian Camporesi–Higuchi Dirac up to a global imaginary scaling that drops out of all load-bearing falsifiers (Paper 43 § 3.2).

2.3 The BBB axiom audit at $(m, n) = (4, 6)$

The Bizi–Brouder–Besnard [3] classification of indefinite spectral triples by $(m,n) = (t+s, t-s)$ mod 8 places Lorentzian $(3,1)$ West-coast at $(m,n) = (4,6)$, with primitive signs (BBB Table 1)

$$\varepsilon = +1, \quad \varepsilon'' = -1, \quad \kappa = -1, \quad \kappa'' = +1. \tag{11}$$

The lift of the Cl(3, 1) charge-conjugation matrix $U_4 = i\gamma^2$ to $\mathcal{K}$ is $J_L = J_{L,\rm spatial} \otimes K_t$ with K_t the identity-unitary times complex conjugation on $\mathbb{C}^{N_t}$ (Paper 43 § 3.3, `geovac/connes_axiom_audit_31.py`). The four BBB-predicted-sign axioms

$$J_L^2 = +I, \quad \{J_L, \gamma^5\} = 0, \quad \{J_L, \eta\} = 0, \quad J_L D_L = +D_L J_L \tag{12}$$

pass bit-exactly (Frobenius residual = 0.0 in float64) at every tested $(n_{\rm max}, N_t) \in \{1,2,3\} \times \{1, 11, 21\}$ (Paper 43 § 3.3). The non-load-bearing universal BBB axiom $\{\gamma^5, D_L\} = 0$ *fails* on truthful $D_{\rm GV}$ for the reason described in Paper 43 § 9.3: this is a known structural feature (chirality-diagonal Camporesi–Higuchi Dirac in the Peskin–Schroeder chiral basis), not a bug, and is preserved by the present construction without affecting any of its load-bearing results.

2.4 Spatial scalar multipliers on the chirality-doubled bundle

The Riemannian Connes–vS multiplier M_{NLM} on $L^2(S^3)$ acts by pointwise multiplication of a scalar hyperspherical harmonic $Y^{(3)}_{NLM}$:

$$\langle (n,l,m_l) \mid M_{NLM} \mid (n',l',m_l') \rangle \;=\; \int_{S^3} \overline{Y_{nlm_l}}\, Y^{(3)}_{NLM}\, Y_{n'l'm_l'}\, dV, \tag{13}$$

with selection rules $|n - n'| + 1 \le N \le n + n' - 1$ (SO(4) triangle), $|l - l'| \le L \le l + l'$ with $l + l' + L$ even (parity), and $M = m_l - m_l'$. The integral is the Avery–Wen–Avery [1] 3-Y integral, evaluated symbolically in exact $\mathbb{Q}$-arithmetic via `geovac/so4_three_y_integral.py`.

The lift to the Weyl-spinor sector $\mathcal{H}_{\rm GV}^{\rm Weyl} \subset \mathcal{H}_{\rm GV}$ is the Clebsch–Gordan-projected matrix

$$\langle (n,l,j,m_j) \mid M_{NLM} \mid (n',l',j',m_j') \rangle \;=\; \sum_{m_l, m_l'} C^{j,m_j}_{l,m_l;1/2,m_s}\, C^{j',m_j'}_{l',m_l';1/2,m_s}\, \delta_{m_s, m_j - m_l}\, (\text{Eq. (13)}), \tag{14}$$

via geovac/spinor_operator_system.py. Both chirality sectors of $\mathcal{H}_{\rm GV} = \mathcal{H}_{\rm GV}^{\rm Weyl} \oplus \mathcal{H}_{\rm GV}^{\rm Weyl}$ see the scalar multiplier identically, so the FullDirac multiplier is

$$M_{NLM}^{\rm spat} \;:=\; M_{NLM}^{\rm Weyl} \oplus M_{NLM}^{\rm Weyl} \;\in\; \mathcal{B}(\mathcal{H}_{\rm GV}^{n_{\max}}), \tag{15}$$

i.e. block-diagonal in the two chirality sectors with equal blocks. This is exactly the FullDiracTruncatedOperato construction of Sprint WH1-R3.5 [28].

2.5 Temporal multipliers on $\mathbb{C}^{N_t}$

We choose the polynomial basis $g_p(t) = t^p$ for $p = 0, 1, \ldots, N_t - 1$, evaluated on the temporal grid (8). The corresponding diagonal matrix is

$$M_p^{\rm temp} \;:=\; \mathrm{diag}\big(g_p(t_0), g_p(t_1), \ldots, g_p(t_{N_t-1})\big) \;\in\; \mathcal{B}(\mathbb{C}^{N_t}). \tag{16}$$

The Vandermonde matrix $V_{kp} = t_k^p$ is invertible at N_t distinct grid points, so the N_t-element family $\{M_p^{\rm temp}\}_{p=0}^{N_t-1}$ spans the full N_t-dimensional commutative subalgebra of diagonal matrices on $\mathbb{C}^{N_t}$. At $N_t = 1$, only the constant $g_0(t) = 1$ is available, and $M_0^{\rm temp} = 1$ is the 1×1 identity.

3 The Lorentzian truncated operator system

Definition 3.1 (Lorentzian truncated operator system)**.** At fixed $(n_{\max}, N_t, T_{\max})$, the *Lorentzian truncated operator system* on the Krein space $\mathcal{K}_{n_{\max},N_t}$ is the $\mathbb{C}$-linear span

$$\mathcal{O}^L_{n_{\max},N_t} \;:=\; \mathrm{span}_{\mathbb{C}}\Big\{ M_{NLM}^{\rm spat} \otimes M_p^{\rm temp} \;:\; (N,L,M) \in \mathcal{S}_{\rm spat}(n_{\max}),\;\; 0 \le p \le N_t - 1 \Big\} \;\subset\; \mathcal{B}(\mathcal{K}_{n_{\max},N_t}), \tag{17}$$

where $\mathcal{S}_{\rm spat}(n_{\max})$ is the set of multiplier labels (N, L, M) admitting a nonzero spinor-lift matrix (15) at cutoff $n_{\max}$ (SO(4) triangle, parity, M-conservation, plus the $n_{\max}$-truncation kernel filter $|n - n'| + 1 \le N \le n + n' - 1$ both ways).

The number of spatial multipliers is exactly $\dim_{\mathbb{C}} \mathcal{O}_{n_{\max}} = 1, 14, 55, 140, 285$ at $n_{\max} = 1, 2, 3, 4, 5$ (Paper 32 § III [28]), and the total number of generators of $\mathcal{O}^L$ is $\dim_{\mathbb{C}} \mathcal{O}_{n_{\max}} \cdot N_t$.

Proposition 3.2 (Operator-system axioms)**.** *At every $(n_{\max}, N_t)$ with $N_t \ge 1$:*

(i) $\mathcal{O}^L_{n_{\max},N_t}$ is closed under conjugate transposition $A \mapsto A^\dagger$ on $\mathcal{B}(\mathcal{K})$.

(ii) $I_{\mathcal{K}} \in \mathcal{O}^L_{n_{\max},N_t}$, with $I_{\mathcal{K}}$ realised by the trivial multiplier $M_{0,0,0}^{\rm spat} \otimes M_0^{\rm temp} = I_{\dim_{\rm spat}} \otimes I_{N_t}$.

(iii) $\mathcal{O}^L_{n_{\max},N_t}$ is not *multiplicatively closed for any $n_{\max} \ge 2$ (witness pair, Theorem 7.1, Section 7).*

Proof. For (i), the spatial conjugation (13) satisfies the Wigner relation

$$\overline{Y_{NLM}^{(3)}} \;=\; (-1)^{M_+}\, Y_{N,L,-M}^{(3)}, \tag{18}$$

for a definite sign $(-1)^{M_+}$ depending on the real-spherical convention (we use complex sphericals so $M_+ = M$); the lifted spinor matrix (15) inherits this and $(M_{NLM}^{\rm spat})^\dagger = \pm M_{N,L,-M}^{\rm spat}$. The temporal multiplier $M_p^{\rm temp}$ is real-diagonal hence self-adjoint: $(M_p^{\rm temp})^\dagger = M_p^{\rm temp}$. Tensor products of $*$-closed families are $*$-closed:

$$(M_{NLM}^{\rm spat} \otimes M_p^{\rm temp})^\dagger \;=\; (M_{NLM}^{\rm spat})^\dagger \otimes M_p^{\rm temp} \;=\; \pm\, M_{N,L,-M}^{\rm spat} \otimes M_p^{\rm temp} \;\in\; \mathcal{O}^L,$$

provided $(N, L, -M) \in \mathcal{S}_{\rm spat}(n_{\max})$, which follows from the M-symmetry of the SO(4) selection rules.

For (ii), the trivial multiplier $M^{\mathrm{spat}}_{0,0,0}$ is the $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$-identity (the constant function $Y^{(3)}_{0,0,0}$ acts as multiplication by a nonzero scalar; under the standard normalisation it acts as the identity matrix), and M^{temp}_0 is the $\mathbb{C}^{N_t}$-identity by $g_0(t) \equiv 1$. The tensor $I_{\dim_{\mathrm{spat}}} \otimes I_{N_t} = I_{\mathcal{K}}$.

For (iii) we postpone to Section 7. □

Remark 3.3 (Operator-system terminology)**.** By standard usage [6, 21], an *operator system* is a $*$-closed unital linear subspace of $\mathcal{B}(\mathcal{H})$ (without the requirement of multiplicative closure). Proposition 3.2 establishes that $\mathcal{O}^L_{n_{\max},N_t}$ is an operator system in this sense. The absence of multiplicative closure (item (iii)) is the structural feature that makes the operator-system substrate *strictly weaker* than the C^*-envelope $C^*(\mathcal{O}^L) \subset \mathcal{B}(\mathcal{K})$; the operator-system data is precisely the algebraic content surviving the spectral truncation that the C^*-envelope discards (cf. Connes–vS [6], “if one takes the C^*-algebra generated by PAP one gets a non-informative full matrix algebra”).

4 Riemannian limit at $N_t = 1$

The load-bearing falsifier for the Lorentzian extension is that at $N_t = 1$ — where the temporal slot reduces to a single point and the construction reduces to a static spatial slice — the operator-system substrate must recover the chirality-doubled Riemannian operator system bit-identically.

Theorem 4.1 (Bit-exact Riemannian-limit recovery at $N_t = 1$)**.** *At $N_t = 1$, the Lorentzian truncated operator system reduces to the chirality-doubled Riemannian operator system on $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$:*

$$\mathcal{O}^L_{n_{\max},1} = \mathcal{O}^{\mathrm{Full\,Dirac}}_{n_{\max}} = \left\{M \oplus M \ :\ M \in \mathcal{O}^{\mathrm{Weyl}}_{n_{\max}}\right\} \subset \mathcal{B}(\mathcal{H}^{n_{\max}}_{\mathrm{GV}}). \tag{19}$$

The reduction is bit-exact in float64: for every multiplier label $(N, L, M, p = 0) \in \mathcal{S}_{\mathrm{spat}}(n_{\max}) \times \{0\}$, the Lorentzian multiplier matrix (15) equals the corresponding chirality-doubled spatial multiplier on $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$ with maximum Frobenius residual $= 0.0$ at every tested $n_{\max} \in \{1, 2, 3\}$.

Proof and computational verification. At $N_t = 1$, $M^{\mathrm{temp}}_0 = 1$ is the 1×1 identity scalar, so the tensor product $M^{\mathrm{spat}}_{NLM} \otimes M^{\mathrm{temp}}_0 = M^{\mathrm{spat}}_{NLM}$ is bit-identical to the chirality-doubled spatial multiplier on $\mathcal{H}^{n_{\max}}_{\mathrm{GV}}$. The full list of multiplier labels $\mathcal{S}_{\mathrm{spat}}(n_{\max})$ at $N_t = 1$ matches the FullDiracTruncatedOperatorSystem multiplier list one-for-one (same SO(4) triangle, parity, and M-conservation filters), and the matrix construction calls the same `build_spinor_multiplier_matrix` routine on the same Avery–Wen–Avery 3-Y integrals.

The computational verification is documented in `tests/test_operator_system_lorentzian.py::test_rie` at $n_{\max} \in \{1, 2, 3\}$,

$n_{\max}$	$\dim_{\mathcal{K}}$	# multipliers	$\dim(\mathcal{O}^L)$	$\max\lVert M_{\mathrm{self}} - M_{\mathrm{ref}}\rVert_F$
1	4	1	1	0.0 (exact)
2	16	14	14	0.0 (exact)
3	40	55	55	0.0 (exact)

The residual is exactly 0.0 in float64 (not merely within machine epsilon) because the two constructions call the same underlying matrix-element routine and the 1×1 tensor with identity is the identity bit-operation in IEEE 754. □

Remark 4.2 (Significance of bit-exactness)**.** Bit-exact Riemannian-limit recovery is the natural consistency condition for any Lorentzian extension that purports to reduce to its Riemannian counterpart at static slices. Failure here would indicate a structural incompatibility between the L2-B Krein-space basis convention (Peskin–Schroeder chiral basis at $(m, n) = (4, 6)$) and the FullDirac spinor-bundle basis convention (Paper 32 § III, Sprint WH1-R3.5 [28]). The passing of the falsifier confirms the two conventions are bit-identical at $N_t = 1$, and the Lorentzian construction inherits the Riemannian spinor lift verbatim.

5 Envelope dependence of the propagation number

The propagation number [6] Definition 2.39 of an operator system $\mathcal{O} \subset M_N(\mathbb{C})$ is the smallest $k \in \mathbb{N} \cup \{\infty\}$ such that $\dim_{\mathbb{C}} \mathcal{O}^k = \dim_{\mathbb{C}} \mathrm{env}(\mathcal{O})$, where $\mathcal{O}^k = \mathrm{span}_{\mathbb{C}}\{a_1 \cdot a_2 \cdots a_k : \ a_i \in \mathcal{O}\}$ and $\mathrm{env}(\mathcal{O})$ is the chosen envelope (typically the C^*-envelope, equal to $M_N(\mathbb{C})$ in the Connes–vS Toeplitz setting). On the chirality-doubled spinor bundle there are two natural envelopes: the *Weyl-doubled achievable* envelope and the *full* envelope. The principal result of this paper is that the propagation number distinguishes them.

5.1 The two envelopes

A scalar-multiplier product $M_1 \cdot M_2$ on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ preserves the chirality-block-diagonal structure:

$$(M_1 \oplus M_1) \cdot (M_2 \oplus M_2) \ = \ (M_1 M_2) \oplus (M_1 M_2). \tag{20}$$

The maximal subspace of $\mathcal{B}(\mathcal{H}_{\mathrm{GV}}^{n_{\max}})$ reachable by scalar-multiplier products is therefore the Weyl-doubled subspace

$$\mathcal{V}_{\mathrm{Weyl}} \ := \ \left\{ M_W \oplus M_W \ : \ M_W \in M_{\dim_{\mathrm{Weyl}}}(\mathbb{C}) \right\} \ \cong \ M_{\dim_{\mathrm{Weyl}}}(\mathbb{C}), \tag{21}$$

with $\dim_{\mathbb{C}} \mathcal{V}_{\mathrm{Weyl}} \ = \ \dim_{\mathrm{Weyl}}^2$. Tensored with the commutative temporal subalgebra $\mathcal{D}_{N_t} \subset M_{N_t}(\mathbb{C})$ of dimension N_t, the achievable subspace on $\mathcal{K}_{n_{\max},N_t}$ is

$$\mathcal{V}_{\mathrm{ach}} \ := \ \mathcal{V}_{\mathrm{Weyl}} \otimes \mathcal{D}_{N_t} \ \subset \ \mathcal{B}(\mathcal{K}), \tag{22}$$

of dimension

$$\dim_{\mathbb{C}} \mathcal{V}_{\mathrm{ach}} \ = \ \dim_{\mathrm{Weyl}}^2 \cdot N_t. \tag{23}$$

By contrast, the *full* $\mathcal{B}(\mathcal{K})$ envelope has

$$\dim_{\mathbb{C}} \mathcal{B}(\mathcal{K}_{n_{\max},N_t}) \ = \ \dim_{\mathcal{K}}^2 \ = \ \dim_{\mathrm{spat}}^2 \cdot N_t^2 \ = \ 4 \dim_{\mathrm{Weyl}}^2 \cdot N_t^2, \tag{24}$$

where the factor of 4 comes from $\dim_{\mathrm{spat}} = 2 \dim_{\mathrm{Weyl}}$ on the chirality-doubled bundle.

Definition 5.1 (Envelope-relative propagation number)**.** For envelopes $\mathcal{V} \in \{\mathcal{V}_{\mathrm{ach}}, \mathcal{B}(\mathcal{K})\}$, the propagation number $\mathrm{prop}_{\mathcal{V}}(\mathcal{O}^L)$ is the smallest $k \in \mathbb{N} \cup \{\infty\}$ such that $\dim_{\mathbb{C}} \mathrm{span}_{\mathbb{C}}(\mathcal{O}^L)^k = \dim_{\mathbb{C}} \mathcal{V}$, where $(\mathcal{O}^L)^k = \mathrm{span}_{\mathbb{C}}\{a_1 a_2 \cdots a_k\}$ is the k-fold product space. We say prop $= \infty$ when no finite k reaches $\dim_{\mathbb{C}} \mathcal{V}$.

5.2 Achievable-envelope propagation: prop = 2

Theorem 5.2 (Achievable-envelope propagation)**.** *For every $(n_{\max}, N_t)$ with $n_{\max} \geq 2$ and $N_t \geq 1$:*

$$\mathrm{prop}_{\mathrm{ach}}(\mathcal{O}^L_{n_{\max},N_t}) \ = \ 2. \tag{25}$$

For every $(n_{\max}, N_t)$ with $n_{\max} \geq 2$ and $N_t \geq 1$ at finite cutoff:

$$\mathrm{prop}_{\mathrm{full}}(\mathcal{O}^L_{n_{\max},N_t}) \ = \ \infty. \tag{26}$$

At $n_{\max} = 1$, the spatial slot is trivial (only the identity multiplier survives), and $\mathrm{prop}_{\mathrm{ach}} = \infty$ as well, the trivial case.

Proof of (26). Every product $M_1 \cdot M_2$ of two scalar-multiplier generators is chirality-block-diagonal by (20), so $(\mathcal{O}^L)^k \subset \mathcal{V}_{\mathrm{Weyl}} \otimes \mathrm{span}_{\mathbb{C}}\{M_p \cdot M_{p'} \cdots\}$ for every k. Since $\mathcal{V}_{\mathrm{Weyl}} \subsetneq M_{\dim_{\mathrm{spat}}}(\mathbb{C})$ (the off-block-diagonal chirality-flipping operators are not in $\mathcal{V}_{\mathrm{Weyl}}$), and the temporal C^*-algebra generated by diagonal matrices is $M_{N_t}(\mathbb{C}) \cap \mathcal{D}_{N_t} = \mathcal{D}_{N_t}$ (the commutative subalgebra is its own C^*-algebra image), we have $(\mathcal{O}^L)^k \subset \mathcal{V}_{\mathrm{ach}}$ for every k. Therefore $\dim_{\mathbb{C}}(\mathcal{O}^L)^k \leq \dim_{\mathbb{C}} \mathcal{V}_{\mathrm{ach}} = \dim_{\mathrm{Weyl}}^2 \cdot N_t$, strictly less than $\dim_{\mathbb{C}} \mathcal{B}(\mathcal{K}) = \dim_{\mathcal{K}}^2 = 4 \dim_{\mathrm{Weyl}}^2 \cdot N_t^2$ for any $N_t \geq 1$ and $n_{\max} \geq 2$. So $\mathrm{prop}_{\mathrm{full}}(\mathcal{O}^L) = \infty$. □

Proof of (25), *computational verification.* The Riemannian Paper 32 § III result is $\mathrm{prop}(\mathcal{O}^{\mathrm{Weyl}}_{n_{\max}}) = 2$ for the Weyl-sector operator system on $\mathcal{H}^{\mathrm{Weyl},n_{\max}}_{\mathrm{GV}}$ (Proposition 5.3, see § 10). Under the chirality-doubling map $\mathcal{O}^{\mathrm{Weyl}}_{n_{\max}} \to \mathcal{O}^{\mathrm{Full\,Dirac}}_{n_{\max}}$, $M_W \mapsto M_W \oplus M_W$, the propagation number is preserved with target envelope $\mathcal{V}_{\mathrm{Weyl}}$ (the doubled image space): $(\mathcal{O}^{\mathrm{Full\,Dirac}})^2$ fills $\mathcal{V}_{\mathrm{Weyl}}$ because $(\mathcal{O}^{\mathrm{Weyl}})^2$ fills $M_{\dim_{\mathrm{Weyl}}}(\mathbb{C}) = \mathcal{V}_{\mathrm{Weyl}}$. Tensoring with the commutative $\mathcal{D}_{N_t}$: products of generators $(M^{\mathrm{spat}}_a \otimes M^{\mathrm{temp}}_p) \cdot (M^{\mathrm{spat}}_b \otimes M^{\mathrm{temp}}_{p'}) = (M^{\mathrm{spat}}_a M^{\mathrm{spat}}_b) \otimes (M^{\mathrm{temp}}_p M^{\mathrm{temp}}_{p'})$ generate $(M^{\mathrm{spat}}_a M^{\mathrm{spat}}_b)$-content in $\mathcal{V}_{\mathrm{Weyl}}$ tensored with products $M^{\mathrm{temp}}_p M^{\mathrm{temp}}_{p'} \in \mathcal{D}_{N_t}$. Since $\{M^{\mathrm{temp}}_p\}_{p=0}^{N_t-1}$ already spans $\mathcal{D}_{N_t}$, $(\mathcal{O}^L)^2$ fills $\mathcal{V}_{\mathrm{Weyl}} \otimes \mathcal{D}_{N_t} = \mathcal{V}_{\mathrm{ach}}$.

The computational verification is documented in `tests/test_operator_system_lorentzian.py::test_pro` and `::test_propagation_number_nmax_3_achievable_envelope`. A complete numerical panel is:

$n_{\max}$	N_t	$\dim_{\mathcal{K}}$	$\dim_{\mathrm{Weyl}}$	$\dim_{\mathrm{ach}}$	$\dim(\mathcal{O}^L)$	$\dim((\mathcal{O}^L)^2)$
1	1	4	2	4	1	1
1	3	12	2	12	3	3
2	1	16	8	64	14	**64**
2	3	48	8	192	42	**192**
2	5	80	8	320	70	**320**
3	1	40	20	400	55	**400**

The bolded entries show $\dim((\mathcal{O}^L)^2) = \dim_{\mathrm{ach}}$, i.e. prop $= 2$ at every cell with $n_{\max} \geq 2$. The $n_{\max} = 1$ cells are the trivial case (the spatial slot has only the identity multiplier $(N, L, M) = (0, 0, 0)$, so the operator system is $\mathbb{C} \cdot I_{\mathrm{spat}} \otimes \mathcal{D}_{N_t}$ of dimension N_t, which is closed under multiplication and so has no nontrivial propagation depth to drive). $\square$

Proposition 5.3 (Paper 32 § III propagation, Weyl sector)**.** *For every* $n_{\max} \in \{2, 3, 4\}$, $\mathrm{prop}(\mathcal{O}^{\mathrm{Weyl}}_{n_{\max}}) = 2$ *on the envelope* $\mathrm{env}(\mathcal{O}^{\mathrm{Weyl}}) = M_{\dim_{\mathrm{Weyl}}}(\mathbb{C})$, *with* $\dim(\mathcal{O}^{\mathrm{Weyl}}) < \dim^2_{\mathrm{Weyl}} = \dim(\mathrm{env})$ *strictly.*

Proof. Paper 32 § III, `prop:propagation_number`, computational verification in `tests/test_operator_system` (24 tests passing, ~ 20 s CPU time). The result is invariant under three SO(4) radial-overlap placeholder conventions (Paper 32 § III, discussion after the proposition statement). $\square$

5.3 Why both numbers are real

The discrepancy between $\mathrm{prop}_{\mathrm{ach}} = 2$ and $\mathrm{prop}_{\mathrm{full}} = \infty$ is not a technical glitch: it reflects the fact that scalar multipliers on a chirality-doubled bundle genuinely cannot reach the full $\mathcal{B}(\mathcal{K})$ envelope at any finite k. The two natural targets correspond to two different operational questions:

Achievable. At what depth k does the operator system fill the maximal subspace structurally consistent with the scalar-multiplier construction (chirality-block-diagonal, temporal-commutative)? Answer: $k = 2$.

Full. At what depth k does the operator system reach the full matrix algebra $M_{\dim_{\mathcal{K}}}(\mathbb{C})$? Answer: never. Reaching this envelope would require either $\mathrm{Cl}(3,1)$-gamma-matrix-valued multipliers (chirality-flipping) or a non-commutative temporal multiplier algebra (e.g. ∂_t-extended), both of which are outside the natural scalar-multiplier construction.

The Riemannian Paper 32 § III situation has no analogous distinction: on the scalar Fock-projected S^3 Hilbert space (Paper 32 Definition 3.2), there is no chirality doubling and no temporal factor, so the achievable and full envelopes coincide. The distinction is genuinely introduced by the Lorentzian extension: the chirality doubling enters at the spinor level (Sprint

WH1-R3.5 `full_dirac_operator_system.py`, which the Riemannian Paper 32 § III analysis chose not to invoke for clarity); the temporal factor enters at the L2-B Krein-space level. Both contribute structural blocks to the propagation that are visible in the present construction but invisible in the Riemannian template.

Remark 5.4 (Connes–vS dual operator system as comparison)**.** Connes–vS [6] Proposition 4.3 establishes that the *dual* operator system to the Toeplitz $C(S^1)^{(n)}$, namely $C(\mathbb{Z})^{(n)} \subset M_n(\mathbb{C})$ generated by characteristic functions on the integer lattice, has prop $= \infty$. Our $\mathrm{prop}_{\mathrm{full}} = \infty$ result has a different mechanism (envelope-side, not generator-side), but the qualitative phenomenon — propagation number depends on which natural envelope is targeted — has a published parallel.

6 Trivial Krein-positive restriction

The Krein-positive cone is the (+1)-eigenspace of J_L:

$$\mathcal{K}^+ \;:=\; \{\psi \in \mathcal{K} \;:\; J_L\psi = +\psi\}, \qquad \dim_{\mathbb{C}} \mathcal{K}^+ = \tfrac{1}{2}\dim_{\mathbb{C}} \mathcal{K}. \tag{27}$$

An operator $A \in \mathcal{B}(\mathcal{K})$ *preserves* $\mathcal{K}^+$ iff $A \cdot \mathcal{K}^+ \subset \mathcal{K}^+$, equivalently iff $[J_L, A] = 0$ at the level of the block decomposition $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$. The Krein-positive restriction of $\mathcal{O}^L$ is

$$\mathcal{O}^{L,+} \;:=\; \{A \in \mathcal{O}^L \;:\; [J_L, A] = 0\}. \tag{28}$$

Proposition 6.1 (Triviality of the Krein-positive restriction)**.** *At every* $(n_{\max}, N_t)$ *with* $N_t \geq 1$*,*

$$\mathcal{O}^{L,+}_{n_{\max},N_t} \;=\; \mathcal{O}^{L}_{n_{\max},N_t}. \tag{29}$$

That is, every scalar-multiplier generator in $\mathcal{O}^L$ *already preserves the Krein-positive cone.*

Proof. A generator $M^{\mathrm{spat}}_{NLM} \otimes M^{\mathrm{temp}}_p$ commutes with $J_L = \gamma^0 \otimes I_{N_t}$ iff M^{spat}_{NLM} commutes with γ^0 and M^{temp}_p commutes with I_{N_t}, the latter being automatic.

For the spatial factor, $M^{\mathrm{spat}}_{NLM} = M_W \oplus M_W$ in the chirality decomposition $\mathcal{H}_{\mathrm{GV}} = \mathcal{H}^{\mathrm{Weyl}}_{\mathrm{GV}} \oplus \mathcal{H}^{\mathrm{Weyl}}_{\mathrm{GV}}$. The fundamental symmetry γ^0 acts as the chirality-swap matrix

$$\gamma^0 \;=\; \begin{pmatrix} 0 & I_{\dim_{\mathrm{Weyl}}} \\ I_{\dim_{\mathrm{Weyl}}} & 0 \end{pmatrix}$$

in the same decomposition (Paper 43 § 3.1, Peskin–Schroeder chiral basis). Direct computation:

$$\gamma^0 \cdot (M_W \oplus M_W) \;=\; \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}\begin{pmatrix} M_W & 0 \\ 0 & M_W \end{pmatrix} \;=\; \begin{pmatrix} 0 & M_W \\ M_W & 0 \end{pmatrix},$$

and

$$(M_W \oplus M_W) \cdot \gamma^0 \;=\; \begin{pmatrix} M_W & 0 \\ 0 & M_W \end{pmatrix}\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \;=\; \begin{pmatrix} 0 & M_W \\ M_W & 0 \end{pmatrix},$$

so $[\gamma^0, M^{\mathrm{spat}}_{NLM}] = 0$ exactly. Tensoring with any operator commuting with I_{N_t} (everything), $[J_L, M^{\mathrm{spat}}_{NLM} \otimes M^{\mathrm{temp}}_p] = 0$. The computational verification is documented in `tests/test_operator_system_lore`
at every tested $(n_{\max}, N_t) \in \{(2,1), (2,3), (3,1)\}$, the fraction of Krein-positive-preserving generators is exactly 1.0. □

Remark 6.2 (The Krein-positivity question shifts to the state level)**.** Proposition 6.1 says that on the chirality-doubled scalar-multiplier operator system, Krein positivity is automatic at the level of the operators themselves. The non-trivial Krein-positivity question is therefore not at the operator-substrate level but at the *state* level: which states on $\mathcal{O}^L$ assign positive expectation values to the Krein-positive matrix elements? The Wasserstein–Kantorovich state space on $\mathcal{K}^+$ is the natural setting (cf. [15] § 3, the Riemannian state-space propinquity), and a Lorentzian-propinquity convergence argument (Sprint L3b, see § 9) would presumably work at the state level rather than the operator-system substrate.

A non-trivial Krein-positive restriction at the substrate level would require enlarging $\mathcal{O}^L$ to include chirality-mixing operators (gamma-matrix-valued multipliers) and/or non-commutative temporal operators (e.g., the Lorentzian Dirac D_L itself, which is not in $\mathcal{O}^L$ as constructed). Both are structurally distinct constructions from the natural pointwise-multiplication operator system studied here.

7 The witness pair

Theorem 7.1 (Witness pair for non-multiplicative-closure)**.** *Fix $n_{\max} \geq 2$, $N_t \geq 1$. Let*

$$a \;:=\; M^{\text{spat}}_{N=2,\,L=1,\,M=0} \otimes M^{\text{temp}}_0 \;\in\; \mathcal{O}^L_{n_{\max},N_t}, \tag{30}$$

and $b := a^\dagger$. Then $a, b \in \mathcal{O}^L_{n_{\max},N_t}$ but $ab \notin \mathcal{O}^L_{n_{\max},N_t}$: the least-squares projection residual

$$\frac{\|ab - \mathrm{proj}_{\mathcal{O}^L}(ab)\|_F}{\|ab\|_F}$$

is bounded below by a positive constant uniformly in N_t. The numerical values at the tested cells are:

$n_{\max}$	N_t	*relative Frobenius residual*
2	1	0.3812
2	3	0.3812
2	5	0.3812
3	1	0.3589

The witness pair certifies that $\mathcal{O}^L$ is genuinely an operator system and not a $*$-algebra: it is closed under conjugate transposition and linear combinations but not *under multiplication.*

Proof. At $p = 0$, $M^{\text{temp}}_0 = I_{N_t}$, so $a = M^{\text{spat}}_{2,1,0} \otimes I_{N_t}$ and $ab = aa^\dagger = \big(M^{\text{spat}}_{2,1,0}(M^{\text{spat}}_{2,1,0})^\dagger\big) \otimes I_{N_t}$.

The spatial factor $M^{\text{spat}}_{2,1,0}(M^{\text{spat}}_{2,1,0})^\dagger$ is the Riemannian witness product of Paper 32 § III [28]: raise-then-lower at the top shell $n = n_{\max}$ produces a diagonal correction on the top-shell entries that no scalar multiplier M^{spat}_{NLM} can supply, because the corresponding $n_{\max}+1$ shell is outside the truncation $P_{n_{\max}}$. Paper 32 § III reports a Frobenius residual of $\sim 14.9\%$ at $n_{\max} = 2$ for the Weyl-sector witness; the chirality-doubled FullDirac version inherits the same residual structure with an arithmetic increase due to the doubled spatial dimension, yielding the 0.3812 value at $n_{\max} = 2$ recorded in `debug/data/l3a_1_lorentzian_operator_system.json`. The residual is independent of N_t at this label because $I^2_{N_t} = I_{N_t}$ is itself in $\mathcal{D}_{N_t}$ (the constant temporal multiplier), so the residual is purely spatial.

At $n_{\max} = 3$, the same construction with the same spatial label $(N, L, M) = (2, 1, 0)$ produces a residual of 0.3589: the percentage drops slightly because the truncation reaches one shell higher, partially absorbing the raise-then-lower correction. This is the expected $n_{\max}$-scaling of the boundary deficit. The verification is recorded in `tests/test_operator_system_lorentzian.py::test_witnes` `::test_witness_pair_nmax_2_Nt_3`, and (for $n_{\max} = 3$) `debug/data/l3a_1_lorentzian_operator_system.j` ∎

Remark 7.2 (Memo–JSON discrepancy at $n_{\max} = 3$)**.** The Sprint L3a-1 memo § 5 quotes the witness-pair residual as 0.381 "identical across N_t values" across a panel that includes $n_{\max} = 3, N_t = 1$. The supporting JSON data `debug/data/l3a_1_lorentzian_operator_system.json`, panel cell $(n_{\max}, N_t) = (3, 1)$, records 0.3589 instead. The discrepancy is purely a memo bookkeeping issue; the JSON and tests agree that the $n_{\max} = 2$ cells give 0.3812 and the $n_{\max} = 3$ cell gives 0.3589, with the slightly lower number reflecting the expected boundary-deficit attenuation at the larger Fock cutoff. We report both the $n_{\max} = 2$ (0.3812) and $n_{\max} = 3$ (0.3589) values explicitly in Theorem 7.1 per the JSON, which is the data of record.

8 Connection to the Paper 43 hemispheric wedge

Paper 43 [33] works with the Krein hemispheric wedge

$$W_L \;:=\; P_W^{\mathrm{spat}} \otimes P_{t\geq 0} \;\subset\; \mathcal{B}(\mathcal{K}_{n_{\max},N_t}), \tag{31}$$

where P_W^{spat} is the Paper 42 [32] hemispheric Hopf-axis m_j-reflection projector on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ and $P_{t\geq 0}$ is the diagonal projector on $\mathbb{C}^{N_t}$ selecting $t_k \geq 0$. The wedge-restricted operator system

$$\mathcal{O}^{L,W}_{n_{\max},N_t} \;:=\; \{\, P_{W_L} A P_{W_L} \;:\; A \in \mathcal{O}^L \,\} \tag{32}$$

is a sub-operator-system on the wedge sub-Krein-space $\mathcal{K}_{W_L} := P_{W_L}\mathcal{K} \cap P_{W_L}\mathcal{K}$ of dimension

$$\dim_{\mathbb{C}} \mathcal{K}_{W_L} \;=\; \dim_{\mathbb{C}} \mathcal{K}_W^{\mathrm{spat}} \cdot \dim_{\mathbb{C}} \mathrm{range}(P_{t\geq 0}) \;=\; \tfrac{1}{2}\dim_{\mathrm{spat}} \cdot \lceil N_t/2 \rceil \tag{33}$$

(Paper 43 § 4.1; for the symmetric grid, $\dim_{\mathbb{C}} \mathrm{range}(P_{t\geq 0}) = \lceil N_t/2 \rceil$).

Proposition 8.1 (Wedge-restricted operator-system data)**.** *At every $(n_{\max}, N_t)$ with $n_{\max} \geq 2, N_t \geq 1$, the wedge-restricted operator system $\mathcal{O}^{L,W}_{n_{\max},N_t}$ is a well-defined sub-operator-system on $\mathcal{K}_{W_L}$. The wedge linear-span dimensions at the tested cells are:*

$n_{\max}$	N_t	$\dim_{\mathbb{C}} \mathcal{K}_{W_L}$	$\dim_{\mathbb{C}} \mathcal{O}^{L,W}$	*comment*
2	1	8	10	*wedge linear span exceeds wedge dim*
2	3	16	20	*same*
2	5	24	30	*same*

The wedge linear-span dimension exceeds the wedge Hilbert-space dimension because $\mathcal{O}^{L,W}$ is a subspace of $\mathcal{B}(\mathcal{K}_{W_L}) \cong M_{\dim_{W_L}}(\mathbb{C})$ whose dimension is $\dim^2_{W_L}$ (64 at the $n_{\max} = 2, N_t = 1$ cell), not $\dim_{W_L}$.

Proof. The wedge projection is a direct application of the `LorentzianWedge` construction from `geovac/modular_hamiltonian_lorentzian.py` (Paper 43 § 4.1). The wedge-restricted matrix is computed by index-restriction: $(P_{W_L} A P_{W_L})|_{i,j\in W_L} = A_{i,j}$ for indices i, j in the wedge basis-index set. The linear-span dimension is computed by SVD on the stack of vec-form wedge-restricted multiplier matrices, documented in `tests/test_operator_system_lorentzian.py::test_restrict_to_loren`
□

Remark 8.2 (Paper 43 modular content reads on the wedge-restricted operator system)**.** Paper 43 [33] constructs the wedge KMS state ρ_W^L and the modular Hamiltonians K_L^α and K_L^{TT} at the matrix level on $\mathcal{K}_{W_L}$. The natural operator-system reading of those constructions is at the level of expectation values on $\mathcal{O}^{L,W}_{n_{\max},N_t}$: the cross-witness collapse of Paper 43 § 6 (Cor. 8.1) is the statement that the trace functional $A \mapsto \mathrm{Tr}_{\mathcal{K}_{W_L}}(\rho_W^L \cdot A)$ on $\mathcal{O}^{L,W}$ is identical across all four physical witness instantiations. The present paper's contribution is to identify the carrier of this construction as the wedge restriction of the natural Lorentzian truncated operator system, completing the bottom-up picture.

This identification does *not* immediately imply that the Paper 43 modular flow extends to a flow on the operator system itself (in the sense of an automorphism group of $\mathcal{O}^L$), because $\mathcal{O}^L$ is not multiplicatively closed and "automorphism" does not literally apply. The natural Lorentzian-propinquity-style replacement — a continuous family of $\mathbb{C}$-linear maps preserving the operator-system data — is the natural next direction (Sprint L3b).

9 Open questions and Sprint L3b outlook

This paper closes the operator-system substrate at finite cutoff. The following questions remain open.

9.1 Continuum Lorentzian propinquity convergence

Question. Does the family $\{\mathcal{T}^L_{n_{\max},N_t}\}_{n_{\max},N_t\to\infty}$ converge in some Lorentzian-propinquity-style topology to a continuum limit triple $\mathcal{T}^L_\infty$ on $S^3 \times \mathbb{R}$?

This is the principal open question. The Riemannian template — Paper 38 [29]'s Latrémolière-propinquity convergence on $S^3 = \mathrm{SU}(2)$, Paper 39 [30]'s tensor-product extension, Paper 40 [31]'s universal $4/\pi$ rate constant on compact Lie groups — relies on five lemmas (L1′ chirality-doubled operator system, L2 central spectral Fejér kernel, L3 Lipschitz spinor bound, L4 Berezin reconstruction, L5 Latrémolière propinquity assembly). The $S^3 \times \mathbb{R}$ analog requires the construction of a joint compact $\times$ non-compact Fejér kernel and a Krein-positive propinquity definition, both of which are mathematically open (`debug/l3_scoping_memo.md` § 4, Q1 and Q4). As of May 2026, no published Lorentzian propinquity has been constructed: Latrémolière's 2018 [16] and 2017/2023 [15] works are strictly Riemannian, as are Hekkelman–McDonald [13, 14] and Farsi–Latrémolière [8, 9].

Recommended first move (Sprint L3b). Replace the non-compact $\mathbb{R}_t$ by a compact S^1_T with circumference $T < \infty$, producing a joint compact $\mathrm{SU}(2) \times S^1_T$ group manifold on which Peter–Weyl analysis applies in both factors. The L2 central spectral Fejér kernel then transports via a product construction with an explicit joint rate. This reduces L3b to a tractable 4–8 week sprint (`debug/l3_scoping_memo.md` § 6). The compactified-temporal target is then the natural setting for a Lorentzian-propinquity assembly that inherits all five Paper 38 lemmas factor-by-factor, modulo Krein-positivity conventions at the Berezin reconstruction step.

9.2 Krein-positivity at the state level

Question. What is the natural Wasserstein–Kantorovich state-space distance on the Krein-positive states of $\mathcal{O}^L$, and does it converge under the Sprint L3b construction?

Proposition 6.1 shifts the Krein-positivity question from the operator-substrate to the state level. The natural object is the set of positive linear functionals $\omega : \mathcal{O}^L \to \mathbb{C}$ with $\omega(A^\dagger A) \geq 0$ for $A \in \mathcal{O}^L$ *and* ω respects the Krein indefinite form (i.e. $\omega(J_L A J_L) = \omega(A)$, the natural Krein invariance). The Wasserstein–Kantorovich distance on this state space is then the candidate Lorentzian-propinquity metric. To our knowledge, no construction of such an object appears in the published NCG literature as of May 2026.

9.3 Nieuviarts twist-morphism applicability

Question. Does the Nieuviarts 2025 [19] twist morphism from Riemannian-twisted to pseudo-Riemannian spectral triples apply to $S^3 = \mathrm{SU}(2)$ at KO-dimension 3?

Nieuviarts [19] Definition 2.2 explicitly restricts the twist morphism to even-dimensional manifolds. The author's companion proceedings note [20] acknowledges that the odd-dimensional case is open. The present construction explicitly bypasses this by working directly with the van den Dungen 2016 [26] Proposition 4.1 lift to a Krein space, which has no parity-dimension restriction. If the Nieuviarts construction were extended to odd dimensions, it would provide an alternative substrate for the Sprint L3b program; as it stands, the BBB Krein-construction path of the present paper is the only path available.

9.4 Multi-focal-composition wall

Question. Does the framework's structural-skeleton scope limitation (Paper 42 § 10 O5; CLAUDE.md § 1.7 multi-focal-wall taxonomy; `debug/multifocal_phase_b_synthesis_memo.md`, 2026-05-08) apply to the Lorentzian extension?

This is open and structurally independent of the present paper; the W2a renormalization wall, the W2b cross-manifold wall, and the W3 calibration-data question (the "second packing axiom" problem) are not addressed by Sprint L3 or the present paper. They remain in the same status as the Riemannian closure of Paper 42.

9.5 Higher-rank compact Lie group extension

Question. For compact G in the Paper 40 [31] class ($G \in \{\mathrm{SU}(N), \mathrm{Spin}(n), \mathrm{Sp}(n), G_2, F_4, E_6, E_7, E_8\}$), does the Lorentzian extension $G \times \mathbb{R}$ admit the same operator-system substrate with envelope-dependent propagation?

The construction in the present paper uses only that $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ is a chirality-doubled spinor bundle on a compact group manifold with multiplier action via chirality-block doubling; the same construction transports immediately to higher-rank G with the spinor bundle replaced by the appropriate G-spinor module. The propagation-number invariant $\mathrm{prop}_{\mathrm{ach}} = 2$ should hold at every G by the same chirality-block argument, since the underlying Paper 32 § III prop$= 2$ result extends to higher-rank G via Paper 40 [31]'s L1$'$ lemma. Verification at SU(3), SU(4), Sp(2), G_2 would be a natural ~ 4 week extension of the present module.

9.6 Cross-manifold W2b

Question. Does the Lorentzian extension transport to the cross-manifold target $\mathcal{T}^L_{S^3} \otimes \mathcal{T}^L_{\mathrm{Hardy}(S^5)}$?

The cross-manifold tensor product is the W2b open question of the multi-focal-composition wall taxonomy (CLAUDE.md § 1.7), structurally blocked at the NCG-framework level by the Paper 24 [27] § V Coulomb/HO category mismatch: the Camporesi–Higuchi spinor bundle on S^3 is a second-order Riemannian construction, while the Bargmann–Segal Hardy-space construction on S^5 is a first-order complex-analytic construction. The category mismatch resurfaces at the spectral-triple level (Paper 32 § VIII.C). The present paper does not address W2b; it remains open.

10 Conclusion

We have generalized the Connes–van Suijlekom [6] truncated operator system from the Riemannian round $S^3 = \mathrm{SU}(2)$ Camporesi–Higuchi spectral triple [5] (Paper 32 § III [28]) to the Bizi–Brouder–Besnard [3] Krein spectral triple at signature $(s,t) = (3,1)$ in the Peskin–Schroeder chiral basis at $(m,n) = (4,6)$, with Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t})$ per van den Dungen 2016 [26] Proposition 4.1. The construction reduces bit-identically to the chirality-doubled Riemannian operator system on $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ at $N_t = 1$, confirming compatibility with the Sprint L2-B Krein-space basis convention (Theorem 4.1).

The principal structural finding is the *envelope dependence of the propagation number* (Theorem 5.2). Targeting the Weyl-doubled achievable envelope $\dim^2_{\mathrm{Weyl}} \cdot N_t$, the Lorentzian extension preserves the Connes–vS propagation-2 result verbatim (eq. (25), matching Paper 32 § III, matching the Toeplitz operator system $C(S^1)^{(n)}$ of [6] Proposition 4.2). Targeting the full $\mathcal{B}(\mathcal{K})$ envelope, the propagation number is ∞ (eq. (26)): chirality-doubling and a commutative temporal subalgebra block the scalar-multiplier construction from generating chirality-flipping operators or non-diagonal temporal operators. This envelope-dependence is a real structural feature of the chirality-doubled Lorentzian extension, not visible in the scalar Riemannian template.

A separate substantive finding is that the Krein-positive restriction to the natural sub-operator-system $\mathcal{O}^{L,+}$ equals $\mathcal{O}^L$ itself (Proposition 6.1). Every chirality-doubled scalar multiplier $M \oplus M$ commutes with the chirality-swap fundamental symmetry J_L, so Krein positivity is automatic at the operator-substrate level. The non-trivial Krein-positivity question is at the *state* level, where the Wasserstein–Kantorovich state-space construction on Krein-positive states is the natural setting for a future Lorentzian-propinquity argument.

Honest scope. This paper closes the operator-system substrate at finite cutoff. It does *not* establish:

1. Continuum / GH-limit / Lorentzian-propinquity convergence (open; Sprint L3b is the recommended first move, § 9.1).
2. Krein-positive state-space convergence (open; § 9.2).
3. Non-commutative temporal multipliers (the natural pointwise-multiplication algebra on $\mathbb{R}_t$ is commutative; this is the canonical Connes–vS construction).
4. Gamma-matrix-valued multipliers (which would enlarge the operator system from $C^\infty(S^3 \times \mathbb{R})$ to $C^\infty(S^3 \times \mathbb{R}, \mathrm{Cl}(\mathbb{R}^4))$ and give a non-trivial Krein-positive restriction).

Position in the GeoVac Lorentzian arc. Together with Paper 43 [33] (the four-witness Wick-rotation theorem at the Lorentzian Krein-wedge level) and the prospective Sprint L3b continuum closure, the present paper constitutes the operator-system substrate of the GeoVac Lorentzian extension at the level the framework can currently reach. Paper 43 does not supersede the present paper, nor vice versa: Paper 43 establishes the modular-Hamiltonian content on a hemispheric wedge; the present paper establishes the operator-system content on the full Krein space and its envelope-relative structural invariants. Both are finite-cutoff closures with the multi-month continuum problem named explicitly and left open.

Position against the broader literature. The Connes–van Suijlekom [6] program deferred the GH-convergence question on a non-trivial example to "elsewhere" three times. Paper 38 [29] closed it on the Riemannian $S^3 = \mathrm{SU}(2)$ at the qualitative-rate Latrémolière-propinquity level; Paper 40 [31] closed it on the universal compact-Lie-group family with rate constant $4/\pi$. The Lorentzian analog of Paper 38 (continuum Lorentzian propinquity on $S^3 \times \mathbb{R}$) remains open. The present paper, together with Paper 43 [33], establishes the finite-cutoff substrate from which any such convergence argument would proceed.

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Appendix: Section reference for the § 5 context cross-reference

The Riemannian Paper 32 § III propagation-number result (Proposition 5.3) was established by direct computation in `geovac/operator_system.py` (the `TruncatedOperatorSystem` class). The verification covers $n_{\max} \in \{2, 3, 4\}$ with multiplier counts $\dim_{\mathbb{C}} \mathcal{O}_{n_{\max}} = 14, 55, 140$ and propagation-table

$n_{\max}$	N_{Fock}	N^2	$\dim(\mathcal{O})$	$\dim(\mathcal{O}^2)$	$\dim(\mathcal{O})/N^2$
2	5	25	14	25	0.560
3	14	196	55	196	0.281
4	30	900	140	900	0.156

reproduced from Paper 32 § III, `prop:propagation_number`. The result is robust under three SO(4) radial-overlap convention choices (including the indicator-only convention, which retains only the support pattern of the matrix-element bookkeeping; see `debug/wh1_round2_propagation_memo.md`, 2026-05-03). The present paper's Theorem 5.2 inherits this robustness verbatim: the achievable-envelope propagation number $\text{prop}_{\text{ach}} = 2$ is the same SO(4) selection-rule structural invariant lifted to the chirality-doubled Lorentzian construction.